■ 비행절차업무기준 [별지 제8호서식] <개정 2026. 10. 30.>

비행절차설계 결과보고서

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ㅇㅇ기관
ㅇㅇ부서

절차명(Title of Procedure) :

절차설계 시작일(Task commences date) :

절차설계담당자(Assign Procedure Designer) :

초안 작성일(Initial Design Ready Date):

수정 횟수 및 수정일자(Revised version number and date):

최종안 작성 및 승인일(Final version ready and approval date):

요약(Executive Summary) :

첨부 : 표준화 문서 보고 서식(예시)

1. ILS APPROACH PROCEDURE(CAT I)
2. ILS APPROACH PROCEDURE(CAT II & III)
3. LOC APPROACH PROCEDURE
4. VOR APPROACH PROCEDURE
5. LNAV APPROACH PROCEDURE
6. LNAV/VNAV APPROACH PROCEDURE
7. CONV SID(DEP)
8. RNAV SID(DEP)
9. CONV STAR
10. RNAV STAR
11. CIRCLING APPROACH COMPUTATION
12. CONV ATS ROUTE
13. RNAV ATS ROUTE

1. ILS APPROACH PROCEDURE(CAT I)

국토교통부 Ministry of Land, Infrastructure and Transport	ILS APPROACH PROCEDURE	
1. TITLE OF PROCEDURE	2. RWY NUMBER	3. REVISION DATE
ILS Z RWY 07 CAT I (2.5%)		
4. OPERATING AGENCY	5. NEW OR REVISED	6. APPROVAL DATE

7. AD INFORMATION				
AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR

8. USE RWY INFORMATION				
RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T°/M°)

9. VISUAL AIDS			
APPROACH LIGHTS	VASIS/PAPI	RUNWAY LIGHTS	OTHER LIGHTS
NOTE			

10. USE NAVAID INFORMATION			
ILS RDH (ft)		GP ANGLE (°)	
NAME	LAT/LONG	ELEV (ft)	MAG VAR

11. SPECIFICATION & USE SENSOR			
SPECIFICATION	GNSS	DME/DME	VOR/DME

12. MAP REFERENCE			
PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE

13. PROCEDURE & REVISION SUMMARY

14. PLAN VIEW OPERATIONAL NOTES

15. MISSED APPROACH INSTRUCTION

16. LANDING MINIMA							
CATEGORY		OCA(H)	A	B	C	D	DL
CAT I	FULL						
	ALS INOP						
CAT II							
NOTE							

17. FIX DATA			
TYPE	IDENT	WP/FIX	LAT/LONG
IAF			
IF			
FAP			
THR			
MATF			
MAHF			

18. TRACK & SEGMENT DATA						
TYPE	IDENT	TRK °M (°T)	DiST (NM)	ALT (ft)	DG (%)	SPEED (kt)
IAF						
IF						
FAP						
THR						
MATF						
MAHF						
NOTE						

19. MINIMUM STABLIZATION DISTANCE(MSD) - RNAV SEGMENTS ONLY				
SEGMENT	DESIGN DIST(NM)	MSD REQUIRED(NM)	REMAIN(NM)	CONFIRMATION

20. TURN PARAMETER						
TYPE	IDENT	TURN ANLGE (°)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90°] (NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)

21. CONTROLLING OBSTACLES								
SEGMENT	ID	NAME	CODE	TYPE	ALTITUDE (ft)		LAT/LONG	
Initial								
Intermediate								
Final1)								
Missed APP (Straight)								
Missed APP (Turning)								
VSS								
NOTE								

22. HOLDING INFORMATION								
IDENT	DIRECTION	ID	NAME	LAT/LONG		ELEV (ft)	MOC (ft)	MIN/MAX ALTITUDE (ft)	SPEED (kt)

23. MINIMUM SECTOR ALTITUDE (MSA)								
NAVAID/WPT		TYPE		IDENT				
TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)
NOTE								

24. EXEMPTION CRITERIA			
REQUIRED		REASON	-
NOTE			

25. FLIGHT CHECK INFORMATION		
DATE OF FLIGHT CHECK	REASON	REMARKS

26. CONCLUSION

27. DEVELOPED BY			
NAME		DATE	

28. APPROVED			
NAME		DATE	

29. REMARKS

	AERONAUTICAL DATA TABULATION

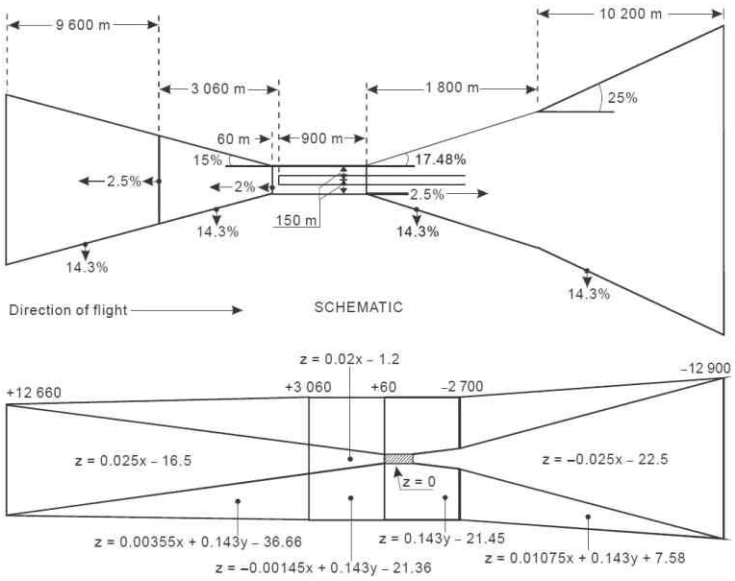
ILS/LOC Approach to RWY 07 from YUMIN(IAF)		
Fix/point		Coordinates

	PRECISION COMPUTATIONS for CAT I (2.5%)

1. COMPUpTATION DATA				
THR ELEV(m)	ILS RDH(m)	LOC THR DIST(m)	GP ANGLE(°)	COURSE WIDTH at THR(m)

MISSED APP CG(%)	AIRCRAFT CATEGORY	DME THR DIST (m)

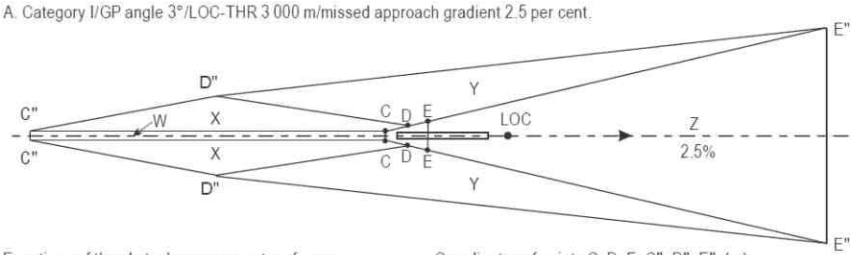
2. OBSTACLE CLEARANCE OF BASIC ILS SURFACE



CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE HEIGHT + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
OBSTACLE LOCATION	
SURFACE HEIGHT AT OBST LOCATION (m)	
PENETRATION (m) : Obstacle height - Surface Height Positive number=Penetration / Negative number=No Penetration	
HEIGHT LOSS MARGIN (m)	
COMPUTATION DH (ft) : MAX [OBST HGT + Hegiht Loss / 200ft]	
COMPUTATION DA (ft) : DH + RWY ELEV	

3. OBSTACLE ASSESSMENT SURFACE(OAS) FOR CAT I

A. Category I/GP angle 3°/LOC-THR 3 000 m/missed approach gradient 2.5 per cent.



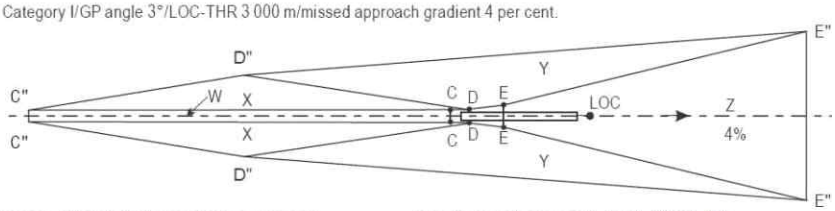
Equations of the obstacle assessment surfaces:

W I z = 0.0285x – 8.01
X I z = 0.027681x + 0.1825y – 16.72
Y I z = 0.023948x + 0.210054y – 21.51
Z I z = –0.025x – 22.50

Coordinates of points C, D, E, C", D", E", (m)

	C	D	E	C"	D"	E"
x	281	-286	-900	10 807	5 438	-12 900
y	49	135	205	96	910	3 001
z	0	0	0	300	300	300

B. Category I/GP angle 3°/LOC-THR 3 000 m/missed approach gradient 4 per cent.



Equations of the obstacle assessment surfaces:

W I z = 0.0285x – 8.01
X I z = 0.027681x + 0.1825y – 16.72
Y I z = 0.020158x + 0.238021y – 26.37
Z I z = –0.04x – 36.00

Coordinates of points C, D, E, C", D", E", (m)

	C	D	E	C"	D"	E"
x	281	-286	-900	10 807	5 438	-8 400
y	49	135	187	96	910	2 082
z	0	0	0	300	300	300

3-1. OAS DATA

APPROACH CATEGORY	GLIDE PATH(°)	LOC THR DIST(m)	RDH(m)	COURSE WIDTH at THR(m)

MISSED APP CG(%)	AIRCRAFT CATEGORY	STANDARD

3-2. FAF or FAP DATA

DISTANCE OF FAF or FAP (NM) : From DME	
FAF or FAP ALTITUDE (m)	
RWY ELEVATION (m)	
FAF or FAP HEIGHT (m)	
INTERMEDIATE MOC (m)	
OAS HEIGHT (m) : FAP - IMT MOC	

3-3. OAS CONSTANTS & TEMPLATE COORDINATES : OAS HEIGHT is 300m

FORMULA		A	B	C
W Surface	Ax+C			
X Surface	Ax+By+C			
Y Surface	Ax+By+C			
Z Surface	Ax+C			

	C"	D"	C	D	E	E""
X						
Y						
Z						

3-4. OAS CONSTANTS & TEMPLATE COORDINATES : OAS HEIGHT is more than 300m

FORMULA		A	B	C
W Surface	Ax+C			
X Surface	Ax+By+C			
Y Surface	Ax+By+C			
Z Surface	Ax+C			

	C"	D"	C	D	E	E""
X						
Y						
Z						

3-5. OAS CONSTANTS & TEMPLATE COORDINATES : Use FAF (descent fix)

FORMULA		A	B	C
W Surface	Ax+C			
X Surface	Ax+By+C			
Y Surface	Ax+By+C			
Z Surface	Ax+C			

	FAF+TOL	C"	D"	C	D	E	E""
X							
Y							
Z							

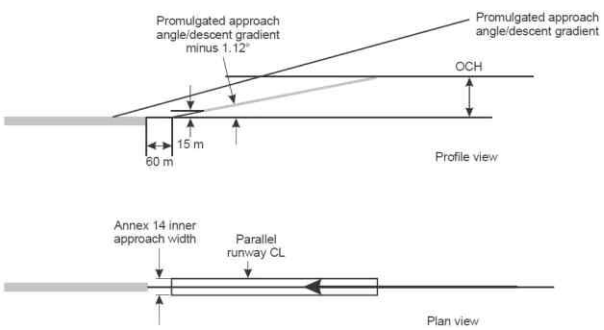
3-6. OBSTACLE CLEARANCE OF OAS - APPROACH PART

CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE HEIGHT + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
OBSTACLE LOCATION	
SURFACE HEIGHT AT OBST LOCATION (m)	
PENETRATION (m) : Obstacle height - Surface Height Positive number=Penetration / Negative number=No Penetration	
HEIGHT LOSS MARGIN (m)	
COMPUTATION DH (ft) : MAX [OBST HGT + Hegiht Loss / 200ft]	
COMPUTATION DA (ft) : DH + RWY ELEV	

3-7. OBSTACLE CLEARANCE OF OAS - MISSED APPROACH PART

CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE HEIGHT + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
OBSTACLE LOCATION	
SURFACE HEIGHT AT OBST LOCATION (m)	
PENETRATION (m) : Obstacle height - Surface Height	

Positive number=Penetration / Negative number=No Penetration		
Z SURFACE HEIGHT (m) : If Obstacle located in Y surface		
Hma (m) Obstacle located in Z surface : Obstacle Height, Obstacle located in Y surface : Z surface Height+Penetration		
EQUIVALENT Ha BY OBSTACLE (m) : $h_a = \frac{h_{ma}\cot Z + (x_o + x)}{\cot Z + \cot\theta}$		
HEIGHT LOSS MARGIN (m)		
COMPUTATION DH (ft) : MAX [Ha + Height Loss / 200ft]		
COMPUTATION DA (ft) : DH + RWY ELEV		
APPROACH DA (ft)		
ADJUSTED DA (ft)		
4. CRM RESULT		
CONTROLLING OBSTACLE (ID/NAME)		
OBSTACLE COORDINATION (LAT/LONG)		
OBSTACLE ELEVATION + TREE + ACCURACY (m)		
RC-X VALUE AT OBSTACLE LOCATION (m)		
RC-Y VALUE AT OBSTACLE LOCATION (m)		
INDIVIDUAL RISK (CATEGORY D/DL)		
MINIMUM DA (ft)		
ADJUSTED DA (ft)		
5. TEMPORARY DA OF PRECISION SEGMENT		
DA OF BASIC ILS SURFACE (ft)		
DA OF OAS (ft)		
DA OF CRM (ft)		
TEMPORARY DA OF PRECISION SEGMENT (ft)		
6. OBSTACLE CLEARANCE OF MISSED APPROACH - After OAS		
CONTROLLING OBSTACLE (ID/NAME)		
OBSTACLE COORDINATION (LAT/LONG)		
OBSTACLE ELEVATION + TREE + ACCURACY (ft)		
OBSTACLE LOCATION-1 (STRAIGHT / TURNING)		
OBSTACLE LOCATION-2 (P or S)		
MOC (ft)		
do VALUE (ft) : From SOC to OBST Location		
tanZ VALUE (%)		
APPROACH DA (ft)		
Height Loss (ft)		
STRAIGHT : (DA - HL) + dotanZ (ft) TURNING : (DA - HL) + dotanZ - MOC (ft)		
PENETRATION (ft) OBST ALT - [(DA - HL) + dotanZ] (STARIGHT) OBST ALT - [(DA - HL) + dotanZ - MOC] (TURNING) Positive number=Penetration / Negative number=No Penetration		
ADJUSTED DA (ft) : 2.5% Climb gradient		
PUBLISHED DA/H (ft) FOR 2.5% CG		
7. VSS (VISUAL SEGMENT SURFACE)		



7-1. VSS DIMENSIONS						
Descent Angle (°)			RC-X	RC-Y(+)	RC-Y(-)	Z(m)
Angle of VSS (°)		BEGIN (ORIGIN)				
OCH (ft)		END				

		(TERMINATION)				
7-2. OBSTACLE ASSESSMENT						
CONTROLLING OBSTACLE (ID/NAME)						
OBSTACLE COORDINATION (LAT/LONG)						
OBSTACLE ELEVATION + TREE + ACCURACY (m)						
RC-X VALUE AT OBSTACLE LOCATION (m)						
RC-Y VALUE AT OBSTACLE LOCATION (m)						
SURFACE ELEVATION AT OBST (m) : Surface HGT + THR ELEV						
PENETRATION (m) Positive number=Penetration / Negative number=No Penetration						
8. VISIBILITY WITHOUT APPROACH LIGHTS						
DH (ft)						
APPROACH CATEGORY		A	B	C	D	DL
REQUIRED VIS FOR DH TO RWY THR DISTANCE (m)						
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)						
PUBLISHED VISIBLITY (m) : Highest from above						
9. VISIBILITY APPROACH LIGHTS						
TYPE & LENGTH OF APPROACH LIGHTS						
DH (ft)						
APPROACH CATEGORY		A	B	C	D	DL
LIGHT REDUCTION (m) : FALS : 720m, IALS : 420m, BALS : 210m						
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)						
VISIBLITY (m) : Highest from above						
10. PUBLISHED MINIMA						
CATEGORY		DA(DH)	A	B	C	D	DL
CAT I (CG 2.5%)	FULL						
	ALS INOP						
11. REMARK							

2. ILS APPROACH PROCEDURE(CAT II & III)

국토교통부 Ministry of Land, Infrastructure and Transport	ILS APPROACH PROCEDURE

1. TITLE OF PROCEDURE	2. RWY NUMBER	3. REVISION DATE
ILS Z RWY 07 CAT II		
4. OPERATING AGENCY	**5. NEW OR REVISED**	**6. APPROVAL DATE**

7. AD INFORMATION

AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG		ARP MAG VAR	

8. USE RWY INFORMATION

RWY NO	DIMENSION (m)	LAT/LONG		THR ELEV (ft)	BEARING(T˚/M˚)	

9. VISUAL AIDS

APPROACH LIGHTS	VASIS/PAPI	RUNWAY LIGHTS	OTHER LIGHTS

NOTE	

10. USE NAVAID INFORMATION

ILS RDH (ft)		GP ANGLE (˚)	
NAME	LAT/LONG	ELEV (ft)	MAG VAR

11. SPECIFICATION & USE SENSOR

SPECIFICATION	GNSS	DME/DME	VOR/DME

12. MAP REFERENCE

PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE

13. PROCEDURE & REVISION SUMMARY

14. PLAN VIEW OPERATIONAL NOTES

15. MISSED APPROACH INSTRUCTION

16. LANDING MINIMA

CATEGORY		DA (DH)	A	B	C	D	DL
CAT I	FULL						
	ALS INOP						
CAT II							
NOTE							

17. FIX DATA

TYPE	IDENT	WP/FIX	LAT/LONG
IAF			
IF			
FAP			
THR			
MATF			
MAHF			

18. TRACK & SEGMENT DATA

TYPE	IDENT	TRK ˚M (˚T)	DiST (NM)	ALT (ft)	DG (%)	SPEED (kt)
IAF						
IF						
FAP						
THR						
MATF						
MAHF						
NOTE						

19. MINIMUM STABLIZATION DISTANCE(MSD) - RNAV SEGMENTS ONLY

SEGMENT	DESIGN DIST(NM)	MSD REQUIRED(NM)	REMAIN(NM)	CONFIRMATION

20. TURN PARAMETER

TYPE	IDENT	TURN ANLGE (˚)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90˚] (NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)

21. CONTROLLING OBSTACLES								
SEGMENT	ID	NAME	CODE	TYPE	ALTITUDE (ft)		LAT/LONG	
Initial								
Intermediate								
Final1)								
Missed APP (Straight)								
Missed APP (Turning)								
VSS								
NOTE								

22. HOLDING INFORMATION								
IDENT	DIRECTION	ID	NAME	LAT/LONG	ELEV (ft)	MOC (ft)	MIN/MAX ALTITUDE (ft)	SPEED (kt)

23. MINIMUM SECTOR ALTITUDE (MSA)								
NAVAID/WPT			TYPE			IDENT		
TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)
NOTE								

24. EXEMPTION CRITERIA			
REQUIRED		REASON	-
NOTE			

25. FLIGHT CHECK INFORMATION		
DATE OF FLIGHT CHECK	REASON	REMARKS

26. CONCLUSION

27. DEVELOPED BY			
NAME		DATE	

28. APPROVED BY			
NAME		DATE	

29. REMARKS

	AERONAUTICAL DATA TABULATION

ILS/LOC Approach to RWY 07 from YUMIN(IAF)		
Fix/point		Coordinates

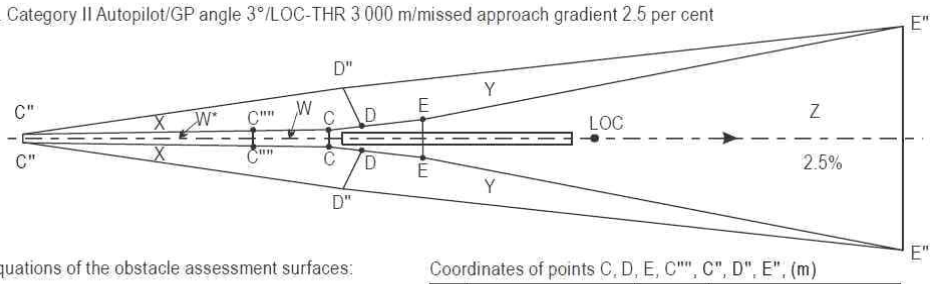
	PRECISION COMPUTATIONS for CAT II & III

1. COMPUTATION DATA				
THR ELEV(m)	ILS RDH(m)	LOC THR DIST(m)	GP ANGLE(°)	COURSE WIDTH at THR(m)

MISSED APP CG(%)	AIRCRAFT CATEGORY	DME THR DIST (m)

2. OBSTACLE ASSESSMENT SURFACE(OAS) FOR CAT II

C. Category II Autopilot/GP angle 3°/LOC-THR 3 000 m/missed approach gradient 2.5 per cent



Equations of the obstacle assessment surfaces:

W l z = 0.0358x – 6.19
W* l z = 0.042x – 12.39
X l z = 0.041370x + 0.2752y – 25.32
Y l z = 0.031955x + 0.280291y – 28.70
Z l z = –0.025x –22.50

Coordinates of points C, D, E, C"", C", D", E", (m)

	C	D	E	C""	C"	D"	E"
x	173	-286	-900	1 000	3 866	-13	-6 900
y	66	135	205	49	55	639	1 424
z	0	0	0	29.6	150	150	150

2-1. OAS DATA

APPROACH CATEGORY	GLIDE PATH(˚)	LOC THR DIST(m)	RDH(m)	COURSE WIDTH at THR(m)

MISSED APP CG(%)	AIRCRAFT CATEGORY	STANDARD

2-2. FAF or FAP DATA

DISTANCE OF FAF or FAP (NM) : From DME	
FAF or FAP ALTITUDE (m)	
RWY ELEVATION (m)	
FAF or FAP HEIGHT (m)	
INTERMEDIATE MOC (m)	
OAS HEIGHT (m) : FAP - IMT MOC	

2-3. OAS CONSTANTS & TEMPLATE COORDINATES

FORMULA		A	B	C
W Surface	Ax+C			
X Surface	Ax+By+C			
Y Surface	Ax+By+C			
Z Surface	Ax+C			

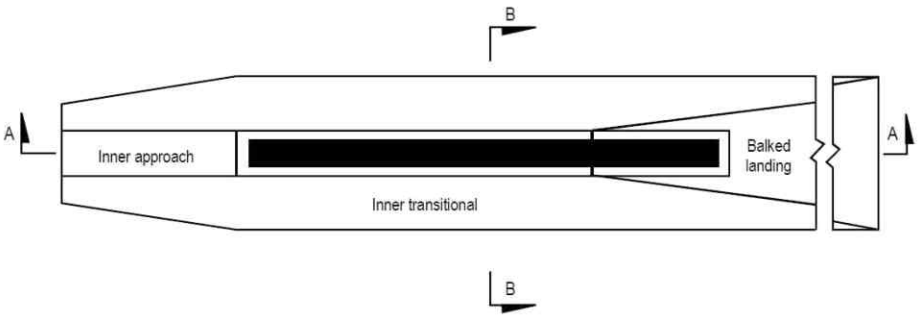
	C"	D"	C	D	E	E""
X						
Y						
Z						

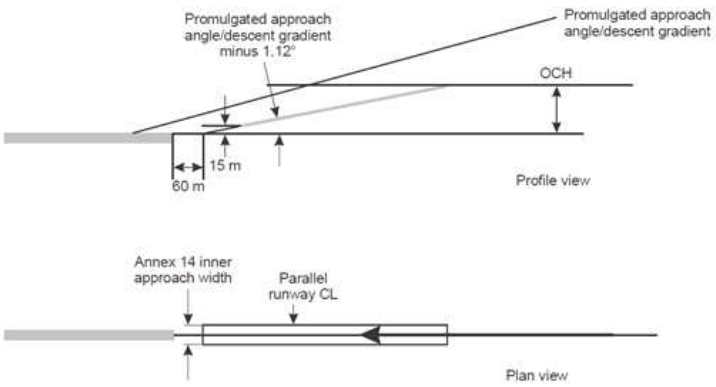
2-4. OBSTACLE CLEARANCE OF OAS - APPROACH PART

CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE HEIGHT + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
OBSTACLE LOCATION	
SURFACE HEIGHT AT OBST LOCATION (m)	
PENETRATION (m) : Obstacle height - Surface Height Positive number=Penetration / Negative number=No Penetration	
HEIGHT LOSS MARGIN (m)	
COMPUTATION DH (ft) : MAX [OBST HGT + Hegiht Loss / 200ft]	
COMPUTATION DA (ft) : DH + RWY ELEV	

2-5. OBSTACLE CLEARANCE OF OAS - MISSED APPROACH PART

CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE HEIGHT + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
OBSTACLE LOCATION	
SURFACE HEIGHT AT OBST LOCATION (m)	
PENETRATION (m) : Obstacle height - Surface Height Positive number=Penetration / Negative number=No Penetration	
Z SURFACE HEIGHT (m) : If Obstacle located in Y surface	
Hma (m) Obstacle located in Z surface : Obstacle Height, Obstacle located in Y surface : Z surface Height+Penetration	
EQUIVALENT Ha BY OBSTACLE (m) : $h_a = \frac{h_{ma}\cot Z + (x_z + x)}{\cot Z + \cot\theta}$	
HEIGHT LOSS MARGIN (m)	

COMPUTATION DH (ft) : MAX [Ha + Height Loss / 200ft]	
COMPUTATION DA (ft) : DH + RWY ELEV	
APPROACH DA (ft)	
ADJUSTED DA (ft)	
3. CRM RESULT	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
INDIVIDUAL RISK (CATEGORY D/DL)	
MINIMUM DA (ft)	
ADJUSTED DA (ft)	
4. TEMPORARY DA OF PRECISION SEGMENT	
DA OF OAS (ft)	
DA OF CRM (ft)	
TEMPORARY DA OF PRECISION SEGMENT (ft)	
5. OBSTACLE CLEARANCE OF MISSED APPROACH - After OAS	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (ft)	
OBSTACLE LOCATION-1 (STRAIGHT / TURNING)	
OBSTACLE LOCATION-2 (P or S)	
MOC (ft)	
do VALUE (ft) : From SOC to OBST Location	
tanZ VALUE (%)	
APPROACH DA (ft)	
Height Loss (ft)	
STRAIGHT : (DA - HL) + dotanZ (ft) TURNING : (DA - HL) + dotanZ - MOC (ft)	
PENETRATION (ft) OBST ALT - [(DA - HL) + dotanZ] (STARIGHT) OBST ALT - [(DA - HL) + dotanZ - MOC] (TURNING) Positive number=Penetration / Negative number=No Penetration	
ADJUSTED DA (ft) : 2.5% Climb gradient	
PUBLISHED DA/H (ft) FOR 2.5% CG	
6. ANNEX 14 OBSTACLE ANALYSIS	
	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE HEIGHT+ TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
OBSTACLE LOCATION	
SURFACE HEIGHT (m)	
PENETRATION (m) : OBST HGT - Surface HGT Positive number=Penetration / Negative number=No Penetration	
7. VSS (VISUAL SEGMENT SURFACE)	



7-1. VSS DIMENSIONS						
Descent Angle (°)			RC-X	RC-Y(+)	RC-Y(-)	Z(m)
Angle of VSS (°)		BEGIN (ORIGIN)				
OCH (ft)		END (TERMINATION)				

7-2. OBSTACLE ASSESSMENT	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
SURFACE ELEVATION AT OBST (m) : Surface HGT + THR ELEV	
PENETRATION (m) Positive number=Penetration / Negative number=No Penetration	

8. VISIBILITY APPROACH LIGHTS (CAT II)						
DH (ft)						
APPROACH CATEGORY	A	B	C	D	DL	
VISIBILITY FROM ICAO DOC 9365 TABLE 6-4 (m)						

9. VISIBILITY APPROACH LIGHTS (CAT III)						
DH (ft)						
APPROACH CATEGORY	A	B	C	D	DL	
VISIBILITY FROM ICAO DOC 9365 TABLE 6-5 (m)						

10. PUBLISHED MINIMA						
CATEGORY	DA(DH)	A	B	C	D	DL
CAT II						
CAT III						

11. REMARKS

3. LOC APPROACH PROCEDURE

국토교통부 Ministry of Land, Infrastructure and Transport	LOC APPROACH PROCEDURE

1. TITLE OF PROCEDURE	2. RWY NUMBER	3. REVISION DATE
LOC Z RWY 07		
4. OPERATING AGENCY	**5. NEW OR REVISED**	**6. APPROVAL DATE**

7. AD INFORMATION

AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR

8. USE RWY INFORMATION

RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T°/M°)

9. VISUAL AIDS

APPROACH LIGHTS	VASIS/PAPI	RUNWAY LIGHTS	OTHER LIGHTS

NOTE	

10. USE NAVAID INFORMATION

ILS RDH (ft)		GP ANGLE (°)	
NAME	LAT/LONG	ELEV (ft)	MAG VAR

11. SPECIFICATION & USE SENSOR

SPECIFICATION	GNSS	DME/DME	VOR/DME

12. MAP REFERENCE

PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE

13. PROCEDURE & REVISION SUMMARY

14. PLAN VIEW OPERATIONAL NOTES

15. MISSED APPROACH INSTRUCTION

16. LANDING MINIMA

CATEGORY		DA (DH)	A	B	C	D	DL
LOC	FULL						
	ALS INOP						

NOTE	

17. RECOMMENDED PROFILE

DME YDM					
ALT(HGT)					

18. FIX DATA

TYPE	IDENT	WP/FIX	LAT/LONG
IAF			
IF			
FAF			
SDF			
MAPt			
THR			
MATF			
MAHF			

19. TRACK & SEGMENT DATA

TYPE	IDENT	TRK °M (°T)	DiST (NM)	ALT (ft)	DG (%)	SPEED (kt)
IAF						
IF						
FAF						
SDF						
MAPt						
THR						
MATF						
MAHF						

NOTE	

20. MINIMUM STABLIZATION DISTANCE(MSD) - RNAV SEGMENTS ONLY

SEGMENT	DESIGN DIST(NM)	MSD REQUIRED(NM)	REMAIN(NM)	CONFIRMATION

21. TURN PARAMETER						
TYPE	IDENT	TURN ANLGE (˚)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90˚] (NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)

22. CONTROLLING OBSTACLES						
SEGMENT	ID	NAME	CODE	TYPE	ALTITUDE (ft)	LAT/LONG
Initial						
Intermediate						
Final (Before SDF)						
Final (After SDF)						
Missed APP (Straight)						
Missed APP (Turning)						
VSS						
NOTE						

23. HOLDING INFORMATION								
IDENT	DIRECTION	ID	NAME	LAT/LONG	ELEV (ft)	MOC (ft)	MIN/MAX ALTITUDE (ft)	SPEED (kt)

24. MINIMUM SECTOR ALTITUDE (MSA)								
NAVAID/WPT		TYPE		IDENT				
TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)
NOTE								

25. EXEMPTION CRITERIA			
REQUIRED		REASON	-
NOTE			

26. FLIGHT CHECK INFORMATION		
DATE OF FLIGHT CHECK	REASON	REMARKS

27. CONCLUSION

28. DEVELOPED BY			
NAME		DATE	

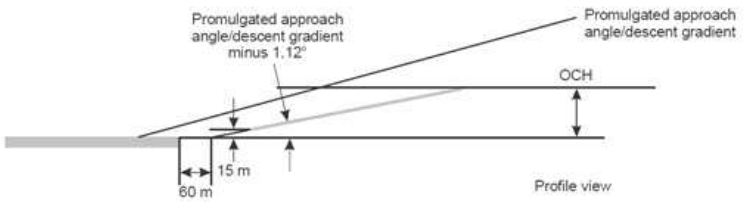
29. APPROVED BY			
NAME		DATE	

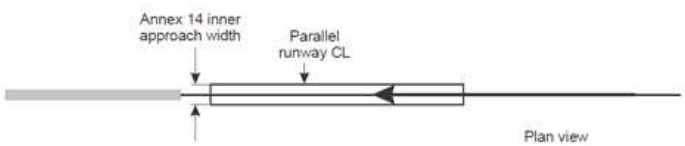
30. REMARKS

	AERONAUTICAL DATA TABULATION
ILS/LOC Approach to RWY 07 from YUMIN(IAF)	

Fix/point		Coordinates

	LOC APPROACH COMPUTATIONS		
NAME OF AIRPORT		INSTRUMENT PROCEDURE	
1. OCA BASED ON FINAL APPROACH SEGMENT			
TYPE OF APPROACH			
FINAL APPROACH CONTROLLING OBSTACLE (ID/NAME)			
OBSTACLE (ft)			
LOCATION (P or S)			
MOC (ft)			
OCA ADJUSTMENTS (ft)			
UNCORRECT ADJUSTED OCA (ft)			
CORRECTED OCA (ft)			
2. OCA BASED ON MISSED APPROACH SEGMENT			
CORRECTED OCA (ft)			
CLIMB GRADIENT (%)			
CONTROLLING OBSTACLE (ID/NAME)			
OBSTACLE ELEVATION (ft)			
OBSTACLE LOCATION-1 : Initial, Intermediate, Final Pahse			
OBSTACLE LOCATION (P or S)			
DISTANCE FROM SOC TO OBST LOCATION (ft) : do			
ESTIMATED AIRCRAFT ALTITUDE AT OBST LOCATION (ft) : OCA + do tanZ			
MOC AT OBSTACLE LOCATION (ft)			
ELEVATION OF SURFACE AT OBSTACLE (ft) : OCA or A/C ALT - MOC			
PENETRATION (ft) : OBST ELEV - SURFACE ELEV Positive number=Penetration / Negative number=No Penetration			
IF PENETRATION EXISTS, ADJUSTED OCA AND/OR ESTABLISH CLIMB GRADIENT			
3. ADJUSTED OCA IF PENETRATION EXISTS			
AMOUNT OF PENETRATION TO BE ADJUSTED (ft)	-		
CORRECTED OCA (ft) : From Item 1			
UNCORRECTED ADJUSTED OCA (ft)			
CORRECTED ADJUSTED OCA (ft)			
4. MISSED APPROACH CLIMB GRADIENT IF PENETRATION EXISTS			
AMOUNT OF PENETRATION TO BE ADJUSTED (ft)	-		
CORRECTED OCA (ft) : From Item 1			
MOC (ft)			
DISTANCE FROM SOC TO OBSTACLE (ft) : do			
REQUIRED CLIMB GRADIENT (%) (OBST ELEV + MOC - OCA) / DISTANCE			
PUBLISHED CLIMB GRADIENT			
5. VSS (VISUAL SEGMENT SURFACE)			





5-1. VSS DIMENSIONS						
Descent Angle (°)			RC-X	RC-Y(+)	RC-Y(-)	Z(m)
Angle of VSS (°)		BEGIN (ORIGIN)				
OCH (ft)		END (TERMINATION)				
5-2. OBSTACLE ASSESSMENT						

CONTROLLING OBSTACLE (ID/NAME)					
OBSTACLE COORDINATION (LAT/LONG)					
OBSTACLE ELEVATION + TREE + ACCURACY (m)					
RC-X VALUE AT OBSTACLE LOCATION (m)					
RC-Y VALUE AT OBSTACLE LOCATION (m)					
SURFACE ELEVATION AT OBST (m) : Surface HGT + THR ELEV					
PENETRATION (m) Positive number=Penetration / Negative number=No Penetration					
6. VISIBILITY WITHOUT APPROACH LIGHTS					
OCH (ft)					
APPROACH CATEGORY	A	B	C	D	DL
REQUIRED VIS FOR MAPt TO RWY THR DISTANCE (m)					
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)					
PUBLISHED VISIBLITY (m) : Highest from above					
7. VISIBILITY WITH APPROACH LIGHTS					
TYPE & LENGTH OF APPROACH LIGHTS					
OCH (ft)					
APPROACH CATEGORY	A	B	C	D	DLLIGHT REDUCTION (m) : FALS : 720m, IALS : 420m, BALS : 210m160 01600160 01600

4. VOR APPROACH PROCEDURE

국토교통부 Ministry of Land, Infrastructure and Transport	VOR APPROACH PROCEDURE

1. TITLE OF PROCEDURE	2. RWY NUMBER	3. REVISION DATE
VOR RWY 07		
4. OPERATING AGENCY	**5. NEW OR REVISED**	**6. APPROVAL DATE**

7. AD INFORMATION

AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR

8. USE RWY INFORMATION

RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T°/M°)

9. VISUAL AIDS

APPROACH LIGHTS	VASIS/PAPI	RUNWAY LIGHTS	OTHER LIGHTS

NOTE	

10. USE NAVAID INFORMATION

RDH (ft)		GP ANGLE (°)	
NAME	LAT/LONG	ELEV (ft)	MAG VAR

11. SPECIFICATION & USE SENSOR

SPECIFICATION	GNSS	DME/DME	VOR/DME

12. MAP REFERENCE

PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE

13. PROCEDURE & REVISION SUMMARY

14. PLAN VIEW OPERATIONAL NOTES

15. MISSED APPROACH INSTRUCTION

16. LANDING MINIMA

CATEGORY		DA (DH)	A	B	C	D	DL
VOR	FULL						
	ALS INOP						

NOTE	

17. RECOMMENDED PROFILE

Final Approach Gradient 5.34%, 325ft/NM	DME YDM				
	ALT(HGT)				

18. FIX DATA

TYPE	IDENT	WP/FIX	LAT/LONG

19. TRACK & SEGMENT DATA

TYPE	IDENT	TRK °M (°T)	DiST (NM)	ALT (ft)	DG (%)	SPEED (kt)

NOTE	

20. MINIMUM STABLIZATION DISTANCE(MSD) - RNAV SEGMENTS ONLY

SEGMENT	DESIGN DIST(NM)	MSD REQUIRED(NM)	REMAIN(NM)	CONFIRMATION

21. TURN PARAMETER

TYPE	IDENT	TURN ANLGE (°)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90°] (NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)
MATF						

22. CONTROLLING OBSTACLES

SEGMENT	ID	NAME	CODE	TYPE	ALTITUDE (ft)	LAT/LONG

Initial						
Intermediate						
Final						
Missed APP (Straight)						
Missed APP (Turning)						
VSS						

NOTE	

23. HOLDING INFORMATION

IDENT	DIRECTION	ID	NAME	LAT/LONG	ELEV (ft)	MOC (ft)	MIN/MAX ALTITUDE (ft)	SPEED (kt)

24. MINIMUM SECTOR ALTITUDE (MSA)

NAVAID/WPT		TYPE		IDENT	

TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)

NOTE	

25. EXEMPTION CRITERIA

REQUIRED		REASON	-

NOTE	

26. FLIGHT CHECK INFORMATION

DATE OF FLIGHT CHECK		REASON		REMARKS

27. CONCLUSION

28. DEVELOPED BY

NAME		DATE	

29. APPROVED BY

NAME		DATE	

30. REMARKS

AERONAUTICAL DATA TABULATION

VOR Approach to RWY 07 from ANBIN (IAF)

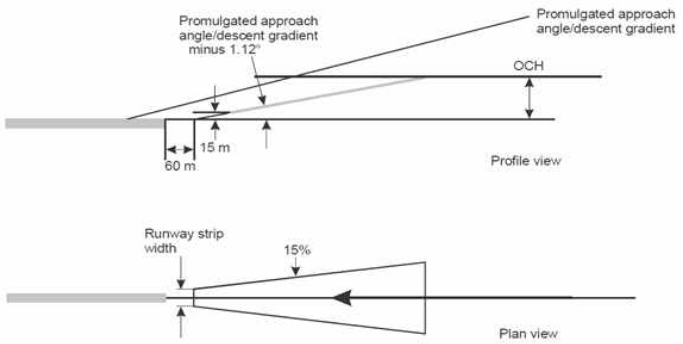
Fix/point		Coordinates

VOR APPROACH COMPUTATIONS

NAME OF AIRPORT		INSTRUMENT PROCEDURE	

1. OCA BASED ON FINAL APPROACH SEGMENT	
TYPE OF APPROACH	
FINAL APPROACH CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE (ft)	
LOCATION (P or S)	
MOC (ft)	
OCA ADJUSTMENTS (ft)	
UNCORRECT ADJUSTED OCA (ft)	
CORRECTED OCA (ft)	
2. OCA BASED ON MISSED APPROACH SEGMENT	
CORRECTED OCA (ft)	
CLIMB GRADIENT (%)	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE ELEVATION (ft)	

OBSTACLE LOCATION-1 : Initial, Intermediate, Final Pahse	
OBSTACLE LOCATION (P or S)	
DISTANCE FROM SOC TO OBST LOCATION (ft) : do	
ESTIMATED AIRCRAFT ALTITUDE AT OBST LOCATION (ft) : OCA + do tanZ	
MOC AT OBSTACLE LOCATION (ft)	
ELEVATION OF SURFACE AT OBSTACLE (ft) : OCA or A/C ALT - MOC	
PENETRATION (ft) : OBST ELEV - SURFACE ELEV Positive number=Penetration / Negative number=No Penetration	
IF PENETRATION EXISTS, ADJUSTED OCA AND/OR ESTABLISH CLIMB GRADIENT	
3. ADJUSTED OCA IF PENETRATION EXISTS	
AMOUNT OF PENETRATION TO BE ADJUSTED (ft)	-
CORRECTED OCA (ft) : From Item 1	
UNCORRECTED ADJUSTED OCA (ft)	
CORRECTED ADJUSTED OCA (ft)	
4. MISSED APPROACH CLIMB GRADIENT IF PENETRATION EXISTS	
AMOUNT OF PENETRATION TO BE ADJUSTED (ft)	-
CORRECTED OCA (ft) : From Item 1	
MOC (ft)	
DISTANCE FROM SOC TO OBSTACLE (ft) : do	
REQUIRED CLIMB GRADIENT (%) (OBST ELEV + MOC - OCA) / DISTANCE	
PUBLISHED CLIMB GRADIENT	
5. VSS (VISUAL SEGMENT SURFACE)	



5-1. VSS DIMENSIONS

Descent Angle (°)			RC-X	RC-Y(+)	RC-Y(-)	Z(m)
Angle of VSS (°)		BEGIN (ORIGIN)				
OCH (ft)		END (TERMINATION)				

5-2. OBSTACLE ASSESSMENT	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
SURFACE ELEVATION AT OBST (m) : Surface HGT + THR ELEV	
PENETRATION (m) Positive number=Penetration / Negative number=No Penetration	

6. VISIBILITY WITHOUT APPROACH LIGHTS					
OCH (ft)					
APPROACH CATEGORY	A	B	C	D	DL
REQUIRED VIS FOR MAPt TO RWY THR DISTANCE (m)					
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)					
PUBLISHED VISIBLITY (m) : Highest from above					
7. VISIBILITY APPROACH LIGHTS					
TYPE & LENGTH OF APPROACH LIGHTS					
OCH (ft)					
APPROACH CATEGORY	A	B	C	D	DL

LIGHT REDUCTION (m) : FALS : 720m, IALS : 420m, BALS : 210m							
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)							
PUBLISHED VISIBLITY (m) : Highest from above							
8. PUBLISHED MINIMA							
CATEGORY		OCA(OCH)	A	B	C	D	DL
VOR	FULL	860(773)					
	ALS INOP						
9. REMARKS							

5. LNAV APPROACH PROCEDURE

국토교통부 Ministry of Land, Infrastructure and Transport	RNAV(GNSS) APPROACH PROCEDURE - LNAV

1. TITLE OF PROCEDURE	2. RWY NUMBER	3. REVISION DATE
RNAV(GNSS) RWY 07 - LNAV		
4. OPERATING AGENCY	**5. NEW OR REVISED**	**6. APPROVAL DATE**

7. AD INFORMATION

AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR

8. USE RWY INFORMATION

RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T˚/M˚)

9. VISUAL AIDS

APPROACH LIGHTS	VASIS/PAPI	RUNWAY LIGHTS	OTHER LIGHTS

NOTE	

10. SPECIFIACATION & USE SENSOR

LNAV RDH (ft)		VPA (˚)	
SPECIFICATION	GNSS	DME/DME	VOR/DME

11. MAP REFERENCE

PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE

12. PROCEDURE & REVISION SUMMARY

13. PLAN VIEW OPERATIONAL NOTES

14. MISSED APPROACH INSTRUCTION

15. LANDING MINIMA

CATEGORY		DA (DH)	A	B	C	D	DL
LNAV	FULL						
	ALS INOP						

NOTE	

16. RECOMMENDED PROFILE

Final Approach Gradient 5.24%, 319ft/NM	FROM RWT						
	ALT(HGT)						

17. FIX DATA

TYPE	IDENT	WP TYPE	LAT/LONG

18. TRACK & SEGMENT DATA

TYPE	IDENT	TRK ˚M (˚T)	DiST (NM)	ALT (ft)	DG (%)	SPEED (kt)

NOTE	

19. MINIMUM STABLIZATION DISTANCE(MSD)

SEGMENT	DESIGN DIST(NM)	MSD REQUIRED(NM)	REMAIN(NM)	CONFIRMATION

20. TURN PARAMETER

TYPE	IDENT	TURN ANLGE (˚)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90˚] (NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)

21. CONTROLLING OBSTACLES

SEGMENT		ID	NAME	CODE	TYPE	ALTITUDE (ft)	LAT/LONG
Initial							
Intermediate							
Final							

(Before SDF)							
Final (After SDF)							
Missed APP (Straight)							
Missed APP (Turning)							
VSS							
NOTE							

22. HOLDING INFORMATION								
IDENT	DIRECTION	ID	NAME	LAT/LONG	ELEV (ft)	MOC (ft)	MIN/MAX ALTITUDE (ft)	SPEED (kt)

23. MINIMUM SECTOR ALTITUDE (MSA)								
NAVAID/WPT		TYPE		IDENT				
TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)
NOTE								

24. EXEMPTION CRITERIA			
REQUIRED		REASON	-
NOTE			

25. FLIGHT CHECK INFORMATION		
DATE OF FLIGHT CHECK	REASON	REMARKS

26. CONCLUSION

27. DEVELOPED BY			
NAME		DATE	

28. APPROVED BY			
NAME		DATE	

29. REMARKS

AERONAUTICAL DATA TABULATION

RNAV(GNSS) RWY 07

Serial Number	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(°T)	Distance (NM)	Turn Direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks
1												
2												
3												

HOLDING PROCEDURE

Holding Identification	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(°T)	Time (min)	Turn Direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks

NONPRECISION COMPUTATIONS - LNAV

NAME OF AIRPORT		INSTRUMENT PROCEDURE	

1. OCA BASED ON FINAL APPROACH SEGMENT

TYPE OF APPROACH	
FINAL APPROACH CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE (ft)	
LOCATION (P or S)	
MOC (ft)	
OCA ADJUSTMENTS (ft)	
UNCORRECT ADJUSTED OCA (ft)	
CORRECTED OCA (ft)	
2. OCA BASED ON MISSED APPROACH SEGMENT	
CORRECTED OCA (ft)	
CLIMB GRADIENT (%)	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE ELEVATION (ft)	
OBSTACLE LOCATION-1 : Initial, Intermediate, Final Pahse	
OBSTACLE LOCATION (P or S)	
DISTANCE FROM SOC TO OBST LOCATION (ft) : do	
ESTIMATED AIRCRAFT ALTITUDE AT OBST LOCATION (ft) : OCA + do tanZ	
MOC AT OBSTACLE LOCATION (ft)	
ELEVATION OF SURFACE AT OBSTACLE (ft) : OCA or A/C ALT - MOC	
PENETRATION (ft) : OBST ELEV - SURFACE ELEV Positive number=Penetration / Negative number=No Penetration	
3. ADJUSTED OCA IF PENETRATION EXISTS	
AMOUNT OF PENETRATION TO BE ADJUSTED (ft)	-
CORRECTED OCA (ft) : From Item 1	
UNCORRECTED ADJUSTED OCA (ft)	
CORRECTED ADJUSTED OCA (ft)	
4. MISSED APPROACH CLIMB GRADIENT IF PENETRATION EXISTS	
AMOUNT OF PENETRATION TO BE ADJUSTED (ft)	-
CORRECTED OCA (ft) : From Item 1	
MOC (ft)	
DISTANCE FROM SOC TO OBSTACLE (ft) : do	
REQUIRED CLIMB GRADIENT (%) (OBST ELEV + MOC - OCA) / DISTANCE	
PUBLISHED CLIMB GRADIENT	
5. VSS (VISUAL SEGMENT SURFACE)	

5-1. VSS DIMENSIONS						
Descent Angle (˚)			RC-X	RC-Y(+)	RC-Y(-)	Z(m)
Angle of VSS (˚)		BEGIN (ORIGIN)				
OCH (ft)		END (TERMINATION)				

5-2. OBSTACLE ASSESSMENT	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
SURFACE ELEVATION AT OBST (m) : Surface HGT + THR ELEV	
PENETRATION (m) Positive number=Penetration / Negative number=No Penetration	

6. VISIBILITY WITHOUT APPROACH LIGHTS					
OCH (ft)					
APPROACH CATEGORY	A	B	C	D	DL
REQUIRED VIS FOR MAPt TO RWY THR DISTANCE (m)					
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)					
PUBLISHED VISIBLITY (m) : Highest from above					

7. VISIBILITY APPROACH LIGHTS	
TYPE & LENGTH OF APPROACH LIGHTS	
OCH (ft)	

APPROACH CATEGORY			A	B	C	D	DL
LIGHT REDUCTION (m) : FALS : 720m, IALS : 420m, BALS : 210m							
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)							
PUBLISHED VISIBLITY (m) : Highest from above							
8. PUBLISHED MINIMA							
CATEGORY		OCA(OCH)	A	B	C	D	DL
LNAV	FULL						
	ALS INOP						
9. REMARKS							

6. LNAV/VNAV APPROACH PROCEDURE

국토교통부 Ministry of Land, Infrastructure and Transport	RNAV(GNSS) APPROACH PROCEDURE - LNAV/VNAV	
1. TITLE OF PROCEDURE	2. RWY NUMBER	3. REVISION DATE
RNAV(GNSS) RWY 07 - LNAV/VNAV		
4. OPERATING AGENCY	5. NEW OR REVISED	6. APPROVAL DATE

7. AD INFORMATION				
AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR

8. USE RWY INFORMATION				
RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T˚/M˚)

9. VISUAL AIDS			
APPROACH LIGHTS	VASIS/PAPI	RUNWAY LIGHTS	OTHER LIGHTS
NOTE			

10. SPECIFIACATION & USE SENSOR			
LNAV/VNAV RDH (ft)		VPA (˚)	
SPECIFICATION	GNSS	DME/DME	VOR/DME

11. MAP REFERENCE			
PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE

12. PROCEDURE & REVISION SUMMARY

13. PLAN VIEW OPERATIONAL NOTES

14. MISSED APPROACH INSTRUCTION

15. LANDING MINIMA							
CATEGORY		DA (DH)	A	B	C	D	DL
LNAV/VNAV	FULL						
	ALS INOP						
NOTE							

16. FIX DATA			
TYPE	IDENT	WP TYPE	LAT/LONG

17. TRACK & SEGMENT DATA						
TYPE	IDENT	TRK ˚M (˚T)	DiST (NM)	ALT (ft)	DG (%)	SPEED (kt)
NOTE						

18. MINIMUM STABLIZATION DISTANCE(MSD)				
SEGMENT	DESIGN DIST(NM)	MSD REQUIRED(NM)	REMAIN(NM)	CONFIRMATION

19. TURN PARAMETER						
TYPE	IDENT	TURN ANLGE (˚)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90˚] (NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)

20. CONTROLLING OBSTACLES									
SEGMENT		ID	NAME	CODE	TYPE	ALTITUDE (ft)		LAT/LONG	
Initial									
Intermediate									
Final (Before SDF)									
Missed									

APP (Straight)									
Missed APP (Turning)									
VSS									
NOTE									

21. TEMPERATURE CORRECTION							
LOW TEMP (˚C)		MIN VPA (˚)		HIGH TEMP (˚C)		MAX VPA (˚)	
YEAR							Average
LOW TEMP OF COLDEST MONTH							
HIGH TEMP OF HOTTEST MONTH							

22. HOLDING INFORMATION								
IDENT	DIRECTION	ID	NAME	LAT/LONG	ELEV (ft)	MOC (ft)	MIN/MAX ALTITUDE (ft)	SPEED (kt)

23. MINIMUM SECTOR ALTITUDE (MSA)								
NAVAID/WPT			TYPE			IDENT		
TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)
NOTE								

24. EXEMPTION CRITERIA			
REQUIRED		REASON	-
NOTE			

25. FLIGHT CHECK INFORMATION		
DATE OF FLIGHT CHECK	REASON	REMARKS

26. CONCLUSION

27. DEVELOPED BY			
NAME		DATE	

28. APPROVED BY			
NAME		DATE	

29. REMARKS

AERONAUTICAL DATA TABULATION

RNAV(GNSS) RWY 07

Serial Number	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(°T)	Distance (NM)	Turn Direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks
1												
2												
3												

HOLDING PROCEDURE

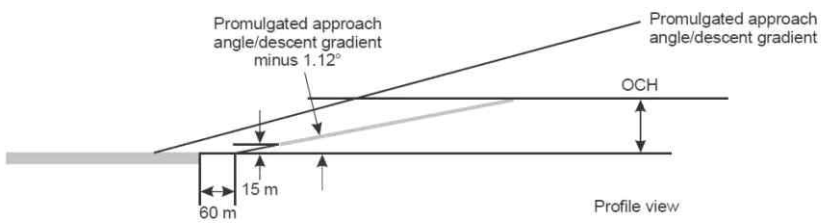
Holding Identification	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(°T)	Time (min)	Turn Direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks

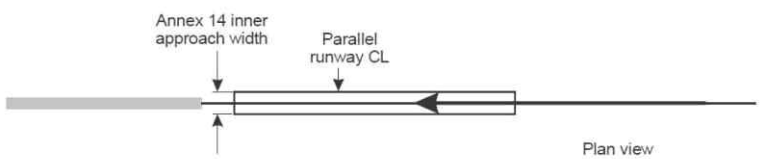
NONPRECISION COMPUTATIONS - LNAV/VNAV

NAME OF AIRPORT		INSTRUMENT PROCEDURE	LNAV/VNAV RWY 07

1. DA BASED ON FINAL APPROACH SEGMENT	
TYPE OF APPROACH	
FINAL APPROACH CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE ELEVATION (ft)	

RWY THRESHOLD ELEVATION (ft)	
OBSTACLE HEIGHT (ft)m	
LOCATION-1	
LOCATION-2 (P or S)	
OAS HEIGHT AT OBSTACLE LOCATION (ft)	
PENETRATION (ft) : OBST ELEV - SURFACE ELEV Positive number=Penetration / Negative number=No Penetration	
HEIGHT LOSS MARGIN (ft)	
UNCORRECT ADJUSTED DH (ft) : OBST HIGT + HL	
MINIMUM DH (ft) : 246ft + HL [DOC 8168 VOL I]	
UNCORRECT ADJUSTED DA (ft)	
CORRECTED DA (ft)	
2. DA BASED ON MISSED APPROACH SEGMENT	
mMISSED APPROACH CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE ELEVATION (ft)	
RWY THRESHOLD ELEVATION (ft)	
OBSTACLE HEIGHT (ft)	
LOCATION-1	
LOCATION-2 (P or S)	
OAS HEIGHT OF PRIMARY AT OBSTACLE LOCATION (ft)	
OAS HEIGHT OF SECONDARY AT OBSTACLE LOCATION (ft)	
PENETRATION (ft) : OBST ELEV - SURFACE ELEV Positive number=Penetration / Negative number=No Penetration	
H'ma (ft) If OBST located in Secondary, OAS HGT of P + Penetration	
ha BY OBSTACLE (ft) :	
HEIGHT LOSS MARGIN (ft)	
UNCORRECT ADJUSTED DH (ft) : OBST HIGT + HL	
MINIMUM DH (ft) : 246ft + HL [DOC 8168 VOL I]	
UNCORRECT ADJUSTED DA (ft)	
CORRECTED DA (ft)	
TEMPERARY PUBLISHED DA (ft) Compare Final with Missed APP DA	
3. DA BASED ON MISSED APPROACH SEGMENT - AFTER OAS	
OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (ft)	
OBSTACLE LOCATION-1 (STRAIGHT / TURNING)	
OBSTACLE LOCATION-2 (P or S)	
MOC (ft)	
do VALUE (ft) : From SOC to OBST Location	
tanZ VALUE (%)	
APPROACH DA (ft)	
Height Loss (ft)	
STRAIGHT : (DA - HL) + dotanZ (ft)	
TURNING : (DA - HL) + dotanZ - MOC (ft)	
PENETRATION (ft) : OBST ALT - [(DA - HL) + dotanZ] (STARIGHT) : OBST ALT - [(DA - HL) + dotanZ - MOC] (TURNING) Positive number=Penetration / Negative number=No Penetration	
IF PENETRATION EXISTS, ADJUSTED DA AND/OR ESTABLISH CLIMB GRADIENT	
4. VSS (VISUAL SEGMENT SURFACE)	





4-1. VSS DIMENSIONS						
Descent Angle (˚)			RC-X	RC-Y(+)	RC-Y(-)	Z(m)
Angle of VSS (˚)		BEGIN (ORIGIN)				
OCH (ft)		END (TERMINATION)				

4-2. OBSTACLE ASSESSMENT	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE COORDINATION (LAT/LONG)	
OBSTACLE ELEVATION + TREE + ACCURACY (m)	
RC-X VALUE AT OBSTACLE LOCATION (m)	
RC-Y VALUE AT OBSTACLE LOCATION (m)	
SURFACE ELEVATION AT OBST (m) : Surface HGT + THR ELEV	
PENETRATION (m) Positive number=Penetration / Negative number=No Penetration	

5. VISIBILITY WITHOUT APPROACH LIGHTS					
OCH (ft)					
APPROACH CATEGORY	A	B	C	D	DL
REQUIRED VIS FOR MAPt TO RWY THR DISTANCE (m)					
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)					
PUBLISHED VISIBLITY (m) : Highest from above					

6. VISIBILITY APPROACH LIGHTS					
TYPE & LENGTH OF APPROACH LIGHTS					
OCH (ft)					
APPROACH CATEGORY	A	B	C	D	DL
LIGHT REDUCTION (m) : FALS : 720m, IALS : 420m, BALS : 210m					
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)					
PUBLISHED VISIBLITY (m) : Highest from above					

7. PUBLISHED MINIMA							
CATEGORY		OCA(OCH)	A	B	C	D	DL
LNAV/VN	FULL						
AV	ALS INOP						

8. REMARKS

7. CONV SID(DEP)

국토교통부 Ministry of Land, Infrastructure and Transport	CONVENTIONAL DEPARTURE PROCEDURE

1. TITLE OF PROCEDURE	2. RWY NUMBER	3. REVISION DATE
4. OPERATING AGENCY	**5. NEW OR REVISED**	**6. APPROVAL DATE**

7. AD INFORMATION

AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR

8. USE RWY INFORMATION

RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T°/M°)

9. USE NAVAID INFORMATION

NAME	LAT/LONG	ELEV (ft)	MAG VAR

10. MAP REFERENCE

PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE

11. PROCEDURE & REVISION SUMMARY

12. PLAN VIEW OPERATIONAL NOTES

13. DEPARTURE ROUTE DESCRIPTION

14. TRACK & SEGMENT DATA

NAME	MAG TRK(°)	DISTANCE (NM)	ALT(ft)	SPEED(kt)	LAT/LONG

NOTE	

15. TURN PARAMETER

IDENT	TURN ANLGE (°)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90°] (NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)

16. CONTROLLING OBSTACLES

ID	NAME	CODE	TYPE	ALTITUDE (ft)	LAT/LONG

NOTE	

17. MINIMUM SECTOR ALTITUDE (MSA)

NAVAID/WPT		TYPE		IDENT	

TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)

NOTE	

18. EXEMPTION CRITERIA

REQUIRED		REASON	-

NOTE	

19. FLIGHT CHECK INFORMATION

DATE OF FLIGHT CHECK	REASON	REMARKS

20. CONCLUSION

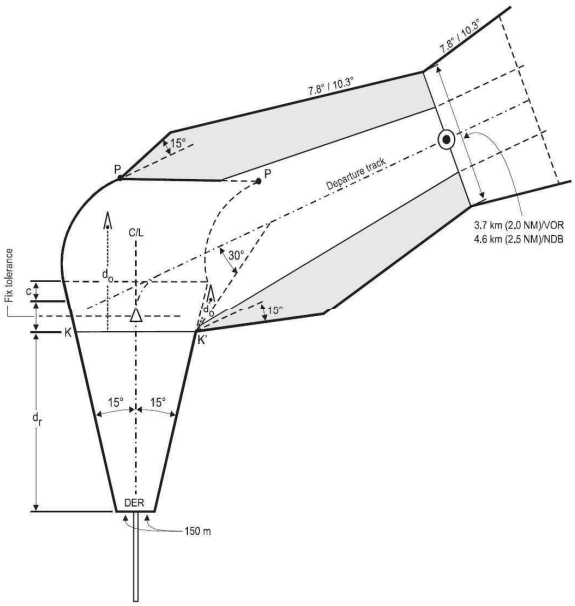
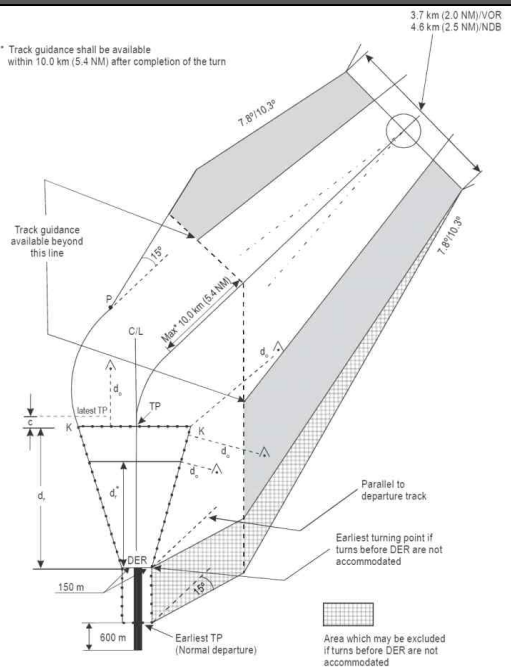
21. DEVELOPED BY

NAME		DATE	

22. APPROVED BY

NAME		DATE	

23. REMARKS

	APPLICATION OF DEPARTURE PROCEDURE		
NAME OF AIRPORT		INSTRUMENT PROCEDURE	
RWY NUMBER & DEPARTURE END OF RUNWAY ELEVATION			
1. EVALUATION RELATIONSHIPS			
			
2. DEPARTURE ROUTE DESCRIPTION			
3. OBSTACLE ANALYSIS BEFORE TURN POINT			
CONTROLLING OBSTACLE (ID/NAME)			
OBSTACLE HEIGHT (ft)			
OIS HEIGHT AT DER (ft) :16ft			
DISTANCE FROM DER TO OBSTACLE (ft)			
MOC (ft) : Distance from DER to OBST * 0.8%			
REQUIRED HEIGHT AT OBST LOCATION (ft) : OBST HGT + MOC			
MINIMUM PDG FOR OBST AVOIDANCE (%) [OBST HGT + MOC - OIS Height at DER] / Distance			
4. OBSTACLE ANALYSIS AFTER TURN POINT			
CONTROLLING OBSTACLE (ID/NAME)			
OBSTACLE HEIGHT (ft)			
OIS HEIGHT AT DER (ft) :16ft			
DISTANCE FROM DER TO KK' LINE : Dr (ft)			
SHORTEST DISTANCE FROM KK' LINE TO OBST : Do (ft)			
① (Dr+Do)*0.8% (ft) : MAX 1000ft (Mountainous area 2000ft)			
② MOC FOR TURNING (ft)			
SELECT MOC (ft) : MAX ①, ②			
REQUIRED HEIGHT AT OBST LOCATION (ft) : OBST HGT + MOC			
MINIMUM PDG FOR OBST AVOIDANCE (%) : [OBST HGT + MOC - OIS Height at DER] / Distance			
5. DETERMINE CLIMB GRADIENT AND ALTITUDE			
CONTROLLING OBSTACLE (ID/NAME)			
OBSTACLE HEIGHT (ft)			
MOC (ft)			
REQUIRED HEIGHT (ft)			
SHORTEST DISTANCE TO OBSTACLE (ft) : Dr+Do			
PUBLISHED PDG (%) : Max PDG			
DIST. FOR OBST AVOIDANCE (ft)			

$\frac{Required\,HGT - 16ft - (Dr + Do) * 3.3\%}{Published\,PDG - 3.3\%}$	
ALTITUDE FOR OBST AVOIDANCE (ft) Distacne for OBST avoidance * Published PDG + 16ft + DER ELEV	
DETERMINE PDG & ALTITUDE	
ATC PURPOSE	
6. CHECK THE LENGTH OF TIA(TURN INITIATION AREA)	
TYPE OF TURN (Turn at FIX, Turn at Altitude)	
MINIMUM HEIGHT OF TURN POINT (ft)	
PUBLISHED PDG (%)	
REQUIRED DISTANCE FROM DER TO TURN POINT (NM)	
DESIGNED DISTANCE FROM DER TO TURN POINT (NM)	
ESTIMATED HEIGHT AT TURN POINT (ft)	
CONFIRMATION	

8. RNAV SID(DEP)

국토교통부 Ministry of Land, Infrastructure and Transport	RNAV DEPARTURE PROCEDURE	
1. TITLE OF PROCEDURE	2. RWY NUMBER	3. REVISION DATE
4. OPERATING AGENCY	5. NEW OR REVISED	6. APPROVAL DATE

7. AD INFORMATION				
AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR

8. USE RWY INFORMATION				
RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T°/M°)

9. SPECIFICATION & USE SENSOR			
SPECIFICATION	GNSS	DME/DME	VOR/DME

10. MAP REFERENCE			
PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE

11. PROCEDURE & REVISION SUMMARY

12. PLAN VIEW OPERATIONAL NOTES

13. DEPARTURE ROUTE DESCRIPTION

14. FIX DATA							
NAME	TYPE	FROM ARP	ATT	XTT	BV	1/2 WIDTH	LAT/LONG

NOTE	

15. TRACK & SEGMENT DATA					
NAME	PATH DESCRIPTOR	TRK °M(°T)	DiST(NM)	ALT(ft)	SPEED(kt)

NOTE	

16. MINIMUM STABLIZATION DISTANCE(MSD)				
SEGMENT	DESIGN DIST(NM)	MSD REQUIRED(NM)	REMAIN(NM)	CONFIRMATION

17. TURN PARAMETER					
IDENT	TURN ANLGE (°)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90°] (NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)

18. CONTROLLING OBSTACLES					
ID	NAME	CODE	TYPE	ALTITUDE (ft)	LAT/LONG

NOTE		

17. MINIMUM SECTOR ALTITUDE (MSA)								
NAVAID/WPT			TYPE			IDENT		
TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)

NOTE	

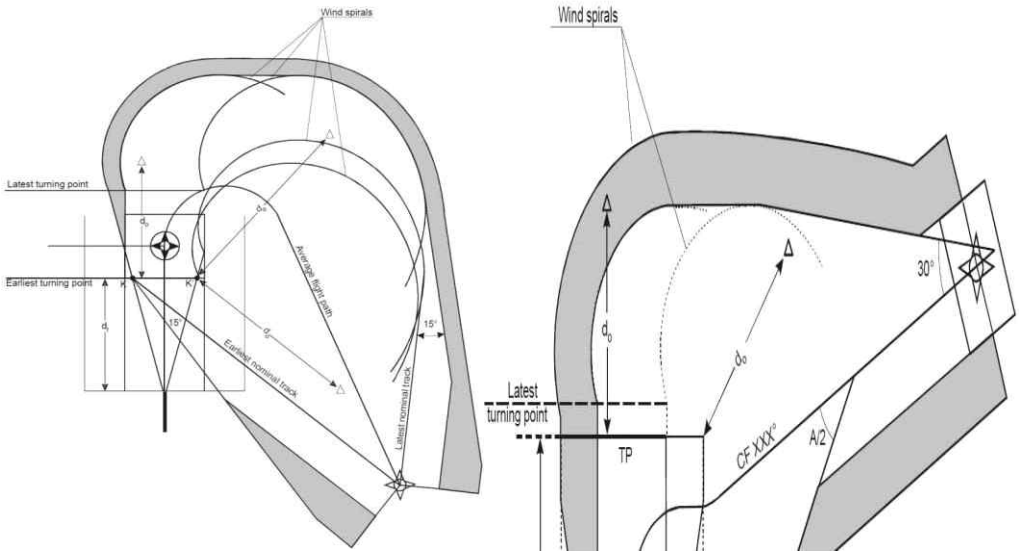
18. EXEMPTION CRITERIA				
REQUIRED		REASON	-	
NOTE				

19. FLIGHT CHECK INFORMATION		
DATE OF FLIGHT CHECK	REASON	REMARKS

20. CONCLUSION		
21. DEVELOPED BY		
NAME		DATE
22. APPROVED BY		
NAME		DATE
23. REMARKS		

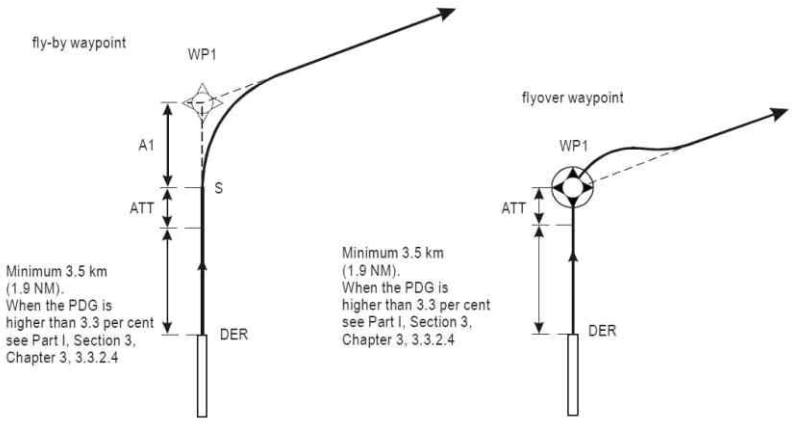
AERONAUTICAL DATA TABULATION												
RNAV KAMIT 2E												
Serial Number	Path Descriptor	Waypoint Identifier	Fly-over	Course /Track °M(°T)	Distance (NM)	Turn Direction	Altitude (ft)	Speed (kt)	Coordinates	VPA /RDH	Navigation specification	Remarks
1												
2												
3												
4												

APPLICATION OF DEPARTURE PROCEDURE			
NAME OF AIRPORT		INSTRUMENT PROCEDURE	
RWY NUMBER & DEPARTURE END OF RUNWAY ELEVATION			
1. EVALUATION RELATIONSHIPS			



2. DEPARTURE ROUTE DESCRIPTION	
3. OBSTACLE ANALYSIS BEFORE TURN POINT	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE HEIGHT (ft)	
OIS HEIGHT AT DER (ft) :16ft	
DISTANCE FROM DER TO OBSTACLE (ft)	
MOC (ft) : Distance from DER to OBST * 0.8%	
REQUIRED HEIGHT AT OBST LOCATION (ft) : OBST HGT + MOC	
MINIMUM PDG FOR OBST AVOIDANCE (%) [OBST HGT + MOC - OIS Height at DER] / Distance	
4. OBSTACLE ANALYSIS AFTER TURN POINT	
CONTROLLING OBSTACLE (ID/NAME)	
OBSTACLE HEIGHT (ft)	
OIS HEIGHT AT DER (ft) :16ft	

DISTANCE FROM DER TO KK' LINE : Dr (ft)		
SHORTEST DISTANCE FROM KK' LINE TO OBST : Do (ft)		
① (Dr+Do)*0.8% (ft) : MAX 1000ft (Mountainous area 2000ft)		
② MOC FOR TURNING (ft)		
SELECT MOC (ft) : MAX ①, ②		
REQUIRED HEIGHT AT OBST LOCATION (ft) : OBST HGT + MOC		
MINIMUM PDG FOR OBST AVOIDANCE (%) : [OBST HGT + MOC - OIS Height at DER] / Distance		
5. DETERMINE CLIMB GRADIENT AND ALTITUDE		
CONTROLLING OBSTACLE (ID/NAME)	BEFORE TURN POINT	AFTER TURN POINT
OBSTACLE HEIGHT (ft)		
MOC (ft)		
REQUIRED HEIGHT (ft)		
SHORTEST DISTANCE TO OBSTACLE (ft) : Dr+Do		
PUBLISHED PDG (%) : Max PDG		
DIST. FOR OBST AVOIDANCE (ft) $\frac{Required\,HGT-16ft-(Dr+Do)*3.3\%}{Published\,PDG-3.3\%}$		
ALTITUDE FOR OBST AVOIDANCE (ft) Distacne for OBST avoidance * Published PDG + 16ft + DER ELEV		
DETERMINE PDG & ALTITUDE		
ATC PURPOSE		
6. CHECK THE LENGTH OF ICA		



TYPE OF TURN (Turn at FIX, Turn at Altitude)	
MINIMUM HEIGHT OF TURN POINT (ft)	
PUBLISHED PDG (%)	
REQUIRED DISTANCE FROM DER TO ETP (NM)	
DESIGNED DISTANCE FROM DER TO ETP (NM)	
ESTIMATED HEIGHT AT TURN POINT (ft)	
CONFIRMATION	

9. CONV STAR

국토교통부 Ministry of Land, Infrastructure and Transport	CONVENTIONAL ARRIVAL PROCEDURE

1. TITLE OF PROCEDURE	2. RWY NUMBER	3. REVISION DATE
4. OPERATING AGENCY	**5. NEW OR REVISED**	**6. APPROVAL DATE**

7. AD INFORMATION

AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR

8. USE RWY INFORMATION

RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T°/M°)

9. USE NAVAID INFORMATION

NAME	LAT/LONG	ELEV (ft)	MAG VAR

10. MAP REFERENCE

PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE

11. PROCEDURE & REVISION SUMMARY

12. PLAN VIEW OPERATIONAL NOTES

13. ARRIVAL ROUTE DESCRIPTION

14. TRACK & SEGMENT DATA

NAME	MAG TRK(°)	DISTANCE (NM)	ALT(ft)	DESCENT GRADIENT (%)	SPEED(kt)	LAT/LONG

NOTE	

15. TURN PARAMETER

IDENT	TURN ANLGE (°)	RADIUS OF TURN[r] (NM)	WIND EFFECT [E90°] (NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)

16. CONTROLLING OBSTACLES

SEGMENT	ID	NAME	CODE	TYPE	ALTITUDE (ft)	LAT/LONG

NOTE	

17. HOLDING INFORMATION

IDENT	DIRECTION	ID	NAME	LAT/LONG	ELEV (ft)	MOC (ft)	MIN/MAX ALTITUDE (ft)	SPEED (kt)

18. MINIMUM SECTOR ALTITUDE (MSA)

NAVAID/WPT		TYPE		IDENT	

TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)

NOTE	

19. EXEMPTION CRITERIA

REQUIRED		REASON	-

NOTE	

20. FLIGHT CHECK INFORMATION

DATE OF FLIGHT CHECK	REASON	REMARKS

21. CONCLUSION

22. DEVELOPED BY

NAME		DATE	

23. APPROVED BY

NAME		DATE	

24. REMARKS

	APPLICATION OF ARRIVAL(CONV) PROCEDURE

NAME OF AIRPORT		INSTRUMENT PROCEDURE	

1. MINIMUM ALTITUDE COMPUTATION

SERIAL NUMBER	1	2	3	4	5	6	7	8	9	10
SEGMENT										
SEGMENT LENGTH (NM)										
COURSE/TRACK°M										
OBS (ID/NAME)										
C O O R D I N A T I O N (LAT/LONG)										
OBS ELEVATION (ft)										
OBS LOCATION (P or S)										
MOC (ft)										
UNCORRECTED MOCA										

(ft)										
PULISHED MOCA (ft)										
ALTITUDE RESTRICTION (ft)										
SPEED RESTRICTION (kt)										

10. RNAV STAR

국토교통부 Ministry of Land, Infrastructure and Transport	RNAV ARRIVAL PROCEDURE

1. TITLE OF PROCEDURE	2. RWY NUMBER	3. REVISION DATE
4. OPERATING AGENCY	**5. NEW OR REVISED**	**6. APPROVAL DATE**

7. AD INFORMATION

AD NAME	ICAO IDENTIFIER	AD ELEV (ft)	ARP LAT/LONG	ARP MAG VAR

8. USE RWY INFORMATION

RWY NO	DIMENSION (m)	LAT/LONG	THR ELEV (ft)	BEARING(T°/M°)

9. SPECIFIACATION & USE SENSOR

SPECIFICATION	GNSS	DME/DME	VOR/DME

10. MAP REFERENCE

PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE

11. PROCEDURE & REVISION SUMMARY

12. PLAN VIEW OPERATIONAL NOTES

13. ARRIVAL ROUTE DESCRIPTION

14. FIX DATA

NAME	TYPE	FROM ARP	ATT	XTT	BV	1/2 WIDTH	LAT/LONG

15. TRACK & SEGMENT DATA

NAME	PATH DESCRIPTOR	TRK °M(°T)	DIST(NM)	ALT(ft)	DESCENT GRADIENT (%)	SPEED(kt)

NOTE

16. MINIMUM STABLIZATION DISTANCE(MSD)

SEGMENT	DESIGN DIST(NM)	MSD REQUIRED(NM)	REMAIN(NM)	CONFIRMATION

17. TURN PARAMETER

IDENT	TURN ANLGE(°)	RADIUS OF TURN[r] (NM)	WINDEFFECT E90°](NM)	EARLIEST TURN POINT (NM)	LATEST TURN POINT (NM)

18. CONTROLLING OBSTACLES

SEGMENT	ID	NAME	CODE	TYPE	ALTITUDE (ft)	LAT/LONG

NOTE

19. HOLDING INFORMATION

IDENT	DIRECTION	ID	NAME	LAT/LONG	ELEV (ft)	MOC (ft)	MIN/MAX ALTITUDE (ft)	SPEED (kt)

20. MINIMUM SECTOR ALTITUDE (MSA)

NAVAID/WPT		TYPE		IDENT	

TO	FROM	ARC (NM)	ID	NAME	ALT (ft)	LAT/LONG	MOC (ft)	MSA (ft)

NOTE

21. EXEMPTION CRITERIA

REQUIRED		REASON	-

NOTE

22. FLIGHT CHECK INFORMATION

DATE OF FLIGHT CHECK	REASON	REMARKS

23. CONCLUSION

24. DEVELOPED BY			
NAME		DATE	

25. APPROVED BY			
NAME		DATE	

26. REMARKS

AERONAUTICAL DATA TABULATION

RNAV(GNSS) RWY 07

Serial Number	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(°T)	Distance (NM)	Turn Direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks
1												
2												
3												
4												

HOLDING PROCEDURE

Holdingldentification	Path Descriptor	Waypoint Identifier	Fly-over	Course/Track °M(°T)	Time (min)	Turn Direction	Altitude (ft)	Speed (kt)	Coordinates	VPA/RDH	Navigation specification	Remarks

APPLICATION OF ARRIVAL(RNAV) PROCEDURE

NAME OF AIRPORT	INCHEON INTL			INSTRUMENT PROCEDURE		GUKDO 1A	
1. MINIMUM ALTITUDE COMPUTATION							
SERIAL NUMBER	1	2	3	4	5	6	7	8
SEGMENT								
SEGMENT LENGTH (NM)								
COURSE/TRACK°M(°T)								
OBS (ID/NAME)								
COORDINATION (LAT/LONG)								
OBS ELEVATION (ft)								
OBS LOCATION (P or S)								
MOC (ft)								
UNCORRECTED MOCA (ft)								
PULISHED MOCA (ft)								
ALTITUDE RESTRICTION (ft)								
SPPED RESTRICTION (kt)								

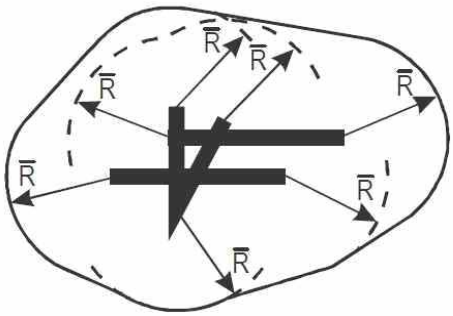
11. CIRCLING APPROACH COMPUTATION

국토교통부 Ministry of Land, Infrastructure and Transport	CIRCLING APPROACH COMPUTATION		
NAME OF AIRPORT		INSTRUMENT PROCEDURE	

1. COMPUTATION DATA			
AERODROME ELEVATION(ft)	TEMPERATURE	WIND FACTOR(kt)	BANK ANGLE(°)

2. RADIUS(R) FOR VISUAL MANOEUVRING (CIRCLING) AREA

Category A aircraft
$\bar{R}$ = 3.16 km
or 1.68 NM
(at 600 m
(2 000 ft) MSL)



CATEGORY OF AIRCRAFT	A	B	C	D
CIRCLING SPEED (kt) : IAS				
AERODROME ELEVATION + 1000ft (ft)				
CIRCLING SPEED (kt) : TAS + 25kt Wind factor				
RADIUS (r) OF TURN (NM) : Using Bank angle 20°				
RADIUS (r) OF TURN (NM) : Using Turn rate of 3° per sec				
STRAIGHT SEGMENT (NM)				
RADIUS (R) FROM THRESHOLD (NM)				

3. OCA/H COMPUTATION				
CATEGORY OF AIRCRAFT	A	B	C	D
CONTROLLING OBSTACLE (ID/NAME)				
OBSTACLE ELEVATION (ft)				
MOC (ft)				
OCA ADJUSTMENTS (ft)				
UNCORRECT ADJUSTED OCA (ft)				
CORRECTED OCA (ft)				
LOWER LIMIT FOR OCA BY PANS-OPS				
OCA FOR STRAIGHT IN APPROACH : LNAV RWY 25				
PUBLISHED OCA (ft)				

4. VISIBILITY COMPUTATION				
CATEGORY OF AIRCRAFT	A	B	C	D
REQUIRED VIS FOR MAPt TO RWY THR DISTANCE (m)				
VISIBILITY FROM ICAO DOC 8168 TABLE I-4-7-3 (m)				
VISIBILITY FROM ICAO DOC 9365 TABLE 6-3 (m)				
VISIBILITY FROM ICAO DOC 9365 TABLE 6-1 (m)				
PUBLISHED VISIBLITY (m) : Highest from above				

5. PUBLISHED MINIMA					
CATEGORY		A	B	C	D
CIRCLING	OCA(H)				
	VISIBILITY				

6. REMARKS

12. CONV ATS ROUTE

국토교통부 Ministry of Land, Infrastructure and Transport	CONVENTIONAL ATS ROUTE

1. ROUTE DESIGNATOR	2. TYPE OF ROUTE	3. REVISION DATE
4. OPERATING AGENCY	**5. NEW OR REVISED**	**6. APPROVAL DATE**

7. NAVIGATIONAL AIDS USED

NAVAID TYPE	ID	FREQ	CHAN	ANT ELEV (ft)

8. MAP REFERENCE

PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE

9. PROCEDURE & REVISION SUMMARY

10. FIX DATA

REFERENCE DATE OF MAG VAR		2015. 01. 01. (WMM 2015)	
SERIAL NUMBER	FIX NAME	LAT/LONG	MAG VAR (°W)

NOTE	

11. TRACK & SEGMENT DATA

SEGMENT	TRK °M-1	TRK °M-2	DIST (NM)	COP	AIRSPACE CLASSIFICATION	MEA (ft)

NOTE	

12-1. PROTECTION AREA (NORMAL TURN METHOD)

FIX	PRIMARY(NM)	BUFFER(NM)	TURN ANGLE(°)	RADIUS OF TURN[r](NM)	VALUE OF T (NM)	EARLIEST LIMIT (NM)	LATEST LIMIT (NM)

12-2. PROTECTION AREA (REFINED TURN METHOD)

FIX	PRIMARY(NM)	BUFFER(NM)	TURN ANGLE(°)	RADIUS OF TURN[r](NM)	WIND EFFECT (E90°)	EARLIEST LIMIT (NM)	LATEST LIMIT (NM)

13. CONTROLLING OBSTACLE DATA

SEGMENT	OBS NAME	CODE	TYPE	ELEV (ft)	LAT/LONG

NOTE	

14. EXEMPTION CRITERIA

REQUIRED		REASON	-

NOTE	

15. FLIGHT CHECK INFORMATION

DATE OF FLIGHT CHECK	REASON	REMARKS

16. DEVELOPED BY			
NAME		DATE	
17. APPROVED BY			
NAME		DATE	
18. REMARKS			

				APPLICATION OF ATS(CONV) ROUTE				
OPERATING AGENCY				INSTRUMENT PROCEDURE				
1. MINIMUM ALTITUDE COMPUTATION								
SERIAL NUMBER	1	2	3	4	5	6	7	8
SEGMENT								
CONTROLLING OBS NAME								
OBS ELEVATION (ft)								
TERRAIN CLASSIFICATION N=Normal, M=Mountainous								
OBS LOCATION (P or S)								
MOC (ft)								
UNCORRECTED MOCA (ft)								
PUBLISHED MOCA (ft)								
ALTITUDE RESTRICTION (ft)								

13. RNAV ATS ROUTE

국토교통부 Ministry of Land, Infrastructure and Transport	RNAV ATS ROUTE

1. ROUTE DESIGNATOR	2. TYPE OF ROUTE	3. REVISION DATE
4. OPERATING AGENCY	**5. NEW OR REVISED**	**6. APPROVAL DATE**

7. SPECIFICATION & SENSOR

SPECIFICATION	GNSS	DME/DME	VOR/DME
RNAV 2			
RNAV 5			

8. MAP REFERENCE

PUBLISHING OFFICE	DATE OF PUBLICATION	MAP NAME	SCALE

9. PROCEDURE & REVISION SUMMARY

10. FIX DATA

REFERENCE DATE OF MAG VAR	2015. 01. 01. (WMM 2015)

SERIAL NUMBER	FIX NAME	VOR/DME BRG & DIST	LAT/LONG	MAG VAR (°W)

NOTE	

11. TRACK & SEGMENT DATA

SEGMENT	NAVIGATION SPECIFICATION	TRK °M-1	TRK °M-2	DIST (NM)	AIRSPACE CLASSIFICATION	MEA (ft)

NOTE	

12. MINIMUM STABILIZATION DISTANCE (MSD)

SEGMENT	DESIGN DIST	MSD REQUIRED	REMAIN	CONFIRMATION

13. PROTECTION AREA

WAYPOINT	SEMI WIDTH (NM)	TURN ANGLE(°)	RADIUS OF TURN[r](NM)	WIND EFFECT (E90°)	EARLIEST POINT (NM)	LATEST POINT (NM)

13. CONTROLLING OBSTACLE DATA

SEGMENT	OBS NAME	CODE	TYPE	ELEV (ft)	LAT/LONG

NOTE	

14. EXEMPTION CRITERIA

REQUIRED		REASON	-

NOTE	

15. FLIGHT CHECK INFORMATION

DATE OF FLIGHT CHECK	REASON	REMARKS

16. DEVELOPED BY

NAME		DATE	

17. APPROVED BY			
NAME		DATE	
18. REMARKS			

	APPLICATION OF ATS(RNAV) ROUTE							
OPERATING AGENCY				INSTRUMENT PROCEDURE				
1. MINIMUM ALTITUDE COMPUTATION								
SERIAL NUMBER	1	2	3	4	5	6	7	8
SEGMENT								
CONTROLLING OBS NAME								
OBS ELEVATION (ft)								
TERRAIN CLASSIFICATION N=Normal, M=Mountainous								
OBS LOCATION (P or S)								
MOC (ft)								
UNCORRECTED MOCA (ft)								
PUBLISHED MOCA (ft)								
PUBLISHED MEA (ft)								